CS315 · ALGORITHMS ANALYSIS & DESIGN

Midterm 451 تدريب

30/10/2023 · ساعة ونص · 25 درجة

محارب

01 عن الميد

ميد الفصل 451 بتاريخ 30/10/2023 للدكتور Mohammed Al-Hagery. المدة ساعة ونص والدرجة من 25. الأسئلة منقولة من صورة الورقة.

Question	1	2	3	4	5	6	Total
Marks	4	4	2	5	4	6	25

مهم

ما عندي نموذج إجابة رسمي لهذا الميد. الحلول من حلي، وكل الجداول انحسبت بكود. أسئلة الـ Matrix chain و LCS و B-tree من محاضرات ما وصلتني سلايداتها، فحليتها بالطريقة المعروفة في الكتاب. لو طريقة الدكتور في السلايدات تختلف، امش على طريقته.

الأسئلة 02

MATRIX CHAIN · 4 PTS **Q1**

Apply the Dynamic programming algorithm and give the optimal solution to the matrix chain multiplication ABCD where the order for matrix.

A = 4 × 5, B = 5 × 8, C = 8 × 3, D = 3 × 7.

Note: Provide the best order of matrix multiplication and the number of operations.

Minimum operations = _____ , Best order of multiplication: _____

	A	B	C	D
A				
B				
C				
D				

HASH TABLE · 4 PTS

Q2

(A): Insert integer keys A = [4, 13, 49, 6, 40, 33] in order into a hash table of size 9 using the hash functions h1(k) = k mod 9. Collisions should be resolved via chaining, where collisions are stored at the end of a chain. Fill in the hash table after all keys have been inserted. [2 pts]

(B): Collisions should be resolved via double hashing, and $h_2(k) = 1 + (k \bmod 5)$. Fill in the hash table after all keys have been inserted. [2 pts]

0	1	2	3	4	5	6	7	8

0	1	2	3	4	5	6	7	8

COUNTING OPERATIONS · 2 PTS

Q3

Count the total primitive operations of the following algorithm.

```
Algorithm arrayMax(A, n)
    currentMax ← A[0]
    Temp ← A[0]
    for i ← 1 to n − 1 do
        if A[i] > currentMax then
            currentMax ← A[i]
    { increment counter i }
    return currentMax
```

TRUE / FALSE + MCQ · 5 PTS **Q4**

Choose the correct answer by selecting True or False:

1) The code length in Huffman codes depend on the frequency of occurrence of characters: a. True b. False

2) Greedy algorithm can solve all the dynamic programming problems. a. True b. False

3) An algorithm which uses the past results and uses them to find new results is a. Greedy Method b. Divide and conquer c. Dynamic programming d. None of the mentioned

4) The time to run MERGESORT on a list of size n, $T(n) = T(\frac{n}{2}) + T(\frac{n}{2}) + T(\frac{MERGE}{2})$ a. True b. False

5) The recursive activity-selection problem algorithm takes a time of O(n log(n)) a. O(n log(n)) b. $O(n^2)$ c. Other value

LCS · 4 PTS

Q5

Using the dynamic programming approach, find the longest common subsequence between two strings X = ACBAED and Y = ABCADF.

	⌀	A	B	C	A	D	F
⌀							
A							
C							
B							
A							
E							
D							

B-TREE · 6 PTS

Q6

a) Create B-Tree of order 5 from the following list of data items: 25, 53, 91, 41, 12, 87, 62, 51 [1 pts]

b) What will the tree look like if we insert key 30? [1 pt]

c) What will the tree look like if we delete keys 53 and 12 from the tree. [2 pts]

d) delete D and then delete X [2 pts]

